

Quiz - Two-species Stability Analysis

My name is: _____

1) Determine the elements of the Community Matrix $\mathbf{A}$ for the two-species model

$$\begin{aligned}\frac{dX}{dt} &= (a - bY)X \\ \frac{dY}{dt} &= (pbX - q)Y,\end{aligned}$$

remembering that, generally-speaking,

$$A_{ij} = \left. \frac{\partial \frac{dx_i}{dt}}{\partial x_j} \right|_{\vec{N}^*}.$$

2) What is the biological interpretation of elements A_{12} and A_{21} ?

3) How would you determine whether a given equilibrium point (not necessarily the coexistence equilibrium) will exhibit the following dynamics after a pulse perturbation?

a) locally stable,

b) locally unstable (including attractor-repeller dynamics),

c) or neutrally stable